

Computer Programming 30509

Introduction to the course
A.Y. 2026/27

Instructors

- **Luca Saglietti (course director)**
- **Paola Leone (Teaching Assistant)**
- **Office hours: I could fix official ones, but it is best (and more flexible) if you just send me an email and we arrange a meeting**
- **Also: USE THE BB FORUM!** (everybody benefits)

Prerequisites

- I'll assume **basic but solid** knowledge of Python (basically the Think Python book, more or less)

Loops, ranges, lists, dictionaries,...

Tools

- We will use Python 3 (I suggest Python 3.6 at least)
- We will also use some scientific libraries (numpy, scipy, matplotlib)
- We will use the Spyder Integrated Development Environment (IDE)
- **All of this comes bundled in the Anaconda distribution**
- During practice sessions, bring your laptop and follow along

Course organization

- **There will be a programming part and a theory part.**
- **They will be tied and proceed in parallel**
- **The theory part is (mostly) intended to introduce concepts that we will then put in practice into code**
- **There might be some tangential topics covered**

This course is still work-in-progress

- **The course has been (and will be) continually adjusted**
 - Based on students' feedback and our observations



Prof. Carlo Baldassi

- **Feedback/constructive criticism is very welcome!**

Course objectives

- **The idea is to learn some more advanced programming patterns/techniques/algorithms, and keep gaining familiarity**
 - Side objective: learn to navigate documentation
- **Also, to cover topics that fit into the course of studies, and prepare for the following semester courses (e.g. machine learning)**
- **Also, to make the topics as interesting/relevant as possible**
- **Also, to have a somewhat coherent framework**

Course topics

- **The main idea is to cover several families of optimization problems, and algorithmic techniques related to them**
- **Then, we will use those as an excuse to learn more advanced programming**
- **Broadly, 4 main topics**
 - Monte Carlo, Simulated Annealing
 - Dynamic programming
 - Non-linear optimization, Gradient descent
 - Data structures (Binary Search Tree)

Course organization (II)

- **For each of the first 3 main topics, we will have the theory part first**
- **Then we will have the programming part related to that (both generic algorithms and examples)**
- **The first two practice sessions will be a bit special because we need some foundations (learn Python's numeric/scientific libraries)**

Practice sessions organization

- **Mostly I will do live-coding explaining what I do and why and how**
- **You're expected to follow along**
- **You're expected to ask questions (this is not the time be shy or passive)**
- **We will also do exercises in class**
- **I'll give exercises for you to do on your own**
- **Feedback is welcome**

Exams

- **Individual project assigned during the mid-term lecture break. Will count for ~5 points**
- **The exam (computer room) will consist in exercises (coding) + some open-ended more theoretically-oriented questions**
- **Exam, 0 to ~26 points, curved on performance**
- **You'll have access to a mock exam and past exams**
- **More about this during the last lecture**

Some general advice

- **Practice, practice, practice**
- **The basics (basic Python + first few lectures) must be rock-solid**
- **Don't fall behind. There is no midterm, you'll need to discipline yourselves**
- **Experiment as much as you can with code**
- **Be curious. Always investigate anything you don't understand 100%. Feel free to ask (lectures, forum, office hours...)**
- **Never look at the solution of an exercise before having given your best**

Some general advice

- **ChatGPT? Claude? Copilot?**

**Questions before
we start?**